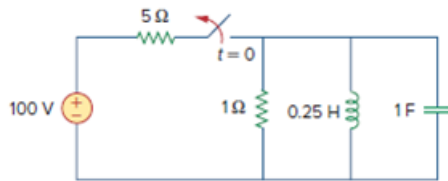


Problem

Find the voltage across the capacitor as a function of time for $t > 0$ for the circuit in Fig. 8.72. Assume steady-state conditions exist at $t = 0^-$.

Figure 8.72:



Step-by-step solution

Step 1 of 8

Initially at $t < 0$, the switch is closed for long time and circuit reaches steady state. Under steady state capacitor acts like an open circuit, inductor acts like a shunted circuit. The equivalent circuit is shown in Figure 1

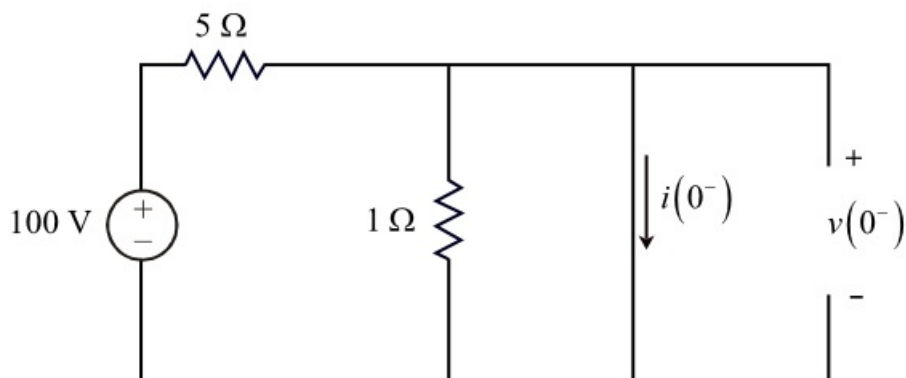


Figure 1

Step 2 of 8

Initial voltage across the capacitor is $v(0^-) = 0 \text{ V}$.

Since voltage across the capacitor cannot change instantaneously. Therefore,

$$\begin{aligned} v(0^+) &= v(0^-) \\ &= 0 \text{ V} \end{aligned}$$

Calculate the current through inductor at $t = 0^-$.

$$\begin{aligned} i(0^-) &= \frac{100 \text{ V}}{5 \Omega} \\ &= 20 \text{ A} \end{aligned}$$

Since, current through inductor cannot change instantaneously. Therefore,

$$\begin{aligned} i(0^+) &= i(0^-) \\ &= 20 \text{ A} \end{aligned}$$

Step 3 of 8

At $t > 0$ the switch is opened. The source is disconnected from the circuit. The equivalent circuit is shown in Figure 2.

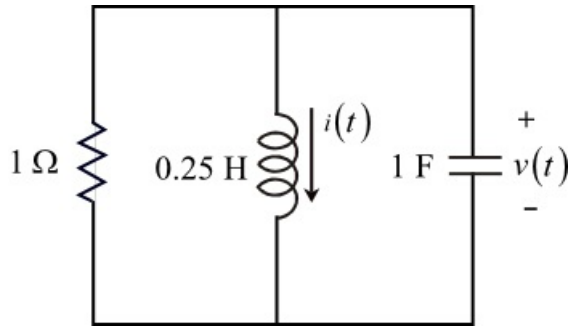


Figure 2

Step 4 of 8

Calculate the neper frequency.

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{(2)(1\ \Omega)(1\ \text{F})} \\ &= 0.5\end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(0.25\ \text{H})(1\ \text{F})}} \\ &= \sqrt{4} \\ &= 2\end{aligned}$$

Calculate the damping frequency.

$$\begin{aligned}\omega_d &= \sqrt{\omega_0^2 - \alpha^2} \\ &= \sqrt{2^2 - 0.5^2} \\ &= \sqrt{3.75} \\ &= 1.936\end{aligned}$$

Step 5 of 8

For $t > 0$ circuit is a source free circuit. The final voltage across the capacitor is zero.

$$V_f = 0\ \text{V}$$

Since $\alpha < \omega_0$, the response is under damped.

Therefore, the voltage response is,

$$v(t) = V_f + e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)\ \text{V}$$

Substitute 0.5 for α , 1.936 for ω_d , and 0 V for V_f in the above equation.

$$\begin{aligned}v(t) &= 0 + e^{-0.5t} [A_1 \cos(1.936t) + A_2 \sin(1.936t)] \\ &= e^{-0.5t} [A_1 \cos(1.936t) + A_2 \sin(1.936t)] \dots\dots (1)\end{aligned}$$

Step 6 of 8

Substitute 0 for t in equation (1).

$$v(0) = e^{-0.5(0)} (A_1 \cos 1.936(0) + A_2 \sin 1.936(0)) \text{ V}$$

$$0 = A_1 \quad (\text{since } v(0) = 0 \text{ V})$$

$$A_1 = 0$$

Apply Kirchhoff's current law at top node of Figure 2.

$$\frac{v(t)}{R} + i(t) + C \frac{dv(t)}{dt} = 0$$

$$C \frac{dv(t)}{dt} = -\frac{v(t)}{R} - i(t)$$

$$\frac{dv(t)}{dt} = -\frac{v(t)}{RC} - \frac{i(t)}{C}$$

$$\frac{dv(t)}{dt} = -\left(\frac{v(t) + Ri(t)}{RC}\right)$$

Substitute 0 for t in the above equation.

$$\frac{dv(0)}{dt} = -\left(\frac{v(0) + Ri(0)}{RC}\right)$$

$$= -\left(\frac{0 + (1)(20)}{(1)(1)}\right)$$

$$= -20 \text{ V/s}$$

Step 7 of 8

Recall equation (1).

$$v(t) = e^{-0.5t} [A_1 \cos(1.936t) + A_2 \sin(1.936t)]$$

Differentiate the above equation with respect to ' t '.

$$\frac{dv(t)}{dt} = e^{-0.5t} \left\{ -0.5 [A_1 \cos(1.936t) + A_2 \sin(1.936t)] + 1.936 [-A_1 \sin(1.936t) + A_2 \cos(1.936t)] \right\}$$

Substitute 0 for t in the above equation.

$$\begin{aligned} \frac{dv(0)}{dt} &= e^{-0.5(0)} \left\{ -0.5 [A_1 \cos(0) + A_2 \sin(0)] + 1.936 [-A_1 \sin(0) + A_2 \cos(0)] \right\} \\ &= e^{-0.5(0)} \{ -0.5 [A_1 + 0] + 1.936 [-0 + A_2] \} \end{aligned}$$

Substitute -20 for $\frac{dv(0)}{dt}$, and 0 for A_1 in the above equation.

$$-20 = -0.5[0 + 0] + 1.936[-0 + A_2]$$

$$\begin{aligned} A_2 &= \frac{-20}{1.936} \\ &= -10.33 \end{aligned}$$

Step 8 of 8

Recall equation (1).

$$v(t) = e^{-0.5t} [A_1 \cos(1.936t) + A_2 \sin(1.936t)]$$

Substitute 0 for A_1 , and -10.33 for A_2 in the above equation.

$$\begin{aligned} v(t) &= e^{-0.5t} [(0) \cos(1.936t) + (-10.33) \sin(1.936t)] \\ &= e^{-0.5t} [0 - 10.33 \sin(1.936t)] \\ &= -10.33e^{-0.5t} \sin(1.936t) \text{ V} \end{aligned}$$

Therefore, the voltage across the capacitor, $v(t)$ for $t > 0$ is $\boxed{-10.33e^{-0.5t} \sin 1.936t \text{ V}}$.